

Jongmin Choi

M.S. Student in Electrical Engineering · Multimodal AI (MMAI) Lab, KAIST

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RESEARCH INTERESTS

Biological intelligence is the one existence proof we have of a general and efficient learner. I believe that networks shaped by its mechanisms rather than by scale alone will prove both stronger and more efficient. My master's work followed this belief along two lines. The first asks how one model should hold several modalities at once, where I studied the gap between audio and text and how to narrow it without flattening what makes each modality distinct. The second asks what the unit of computation should be. A biological neuron is not a perceptron. It fires and resets and carries the imprint of its membrane channels, and I designed spiking neuron algorithms that place these mechanisms inside the unit rather than around it. I am now working on the connector that joins an audio encoder to a language model decoder, where much of the information bottleneck forms, and on the mathematics of optimization for training deep networks.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2025 – Present
M.S. in Electrical Engineering

Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab · GPA 4.06 / 4.30

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2018 – Feb 2025
B.S. in Electrical Engineering and Mathematical Sciences (Double Major)

GPA 3.65 / 4.30

PUBLICATIONS

[C] Conference · [P] Preprint · * Equal contribution

[P] **FiTS: Interpretable Spiking Neurons via Frequency Selectivity and Temporal Shaping**
Jongmin Choi, Joon Son Chung

Submitted to *Conference on Neural Information Processing Systems (NeurIPS)* · 2026 UNDER REVIEW

[C] **LAMB: LLM-based Audio Captioning with Modality Gap Bridging via Cauchy–Schwarz Divergence**
Hyeongkeun Lee*, Jongmin Choi*, KiHyun Nam, Joon Son Chung

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) · 2026 ORAL

[C] **Diffusion-Link: Diffusion Probabilistic Model for Bridging the Audio–Text Modality Gap**
KiHyun Nam*, Jongmin Choi*, Hyeongkeun Lee, Jungwoo Heo, Joon Son Chung

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) · 2026 POSTER

[P] **Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models**
Chaeyoung Jung*, Youngjoon Jang*, Jongmin Choi, Joon Son Chung

arXiv preprint · 2025

TEACHING EXPERIENCE

EE.70038 — Speech Recognition Systems — KAIST Spring 2026
Teaching Assistant · Instructor: **Prof. Joon Son Chung**

EE.40034 — Deep Learning for Visual Understanding — KAIST Fall 2025
Teaching Assistant · Instructor: **Prof. Joon Son Chung**

SKILLS

Research & ML: Python, PyTorch · **iOS Development:** Swift, SwiftUI, SwiftData · **Languages:** Korean (native), English (professional working proficiency)

Last updated: August 2026